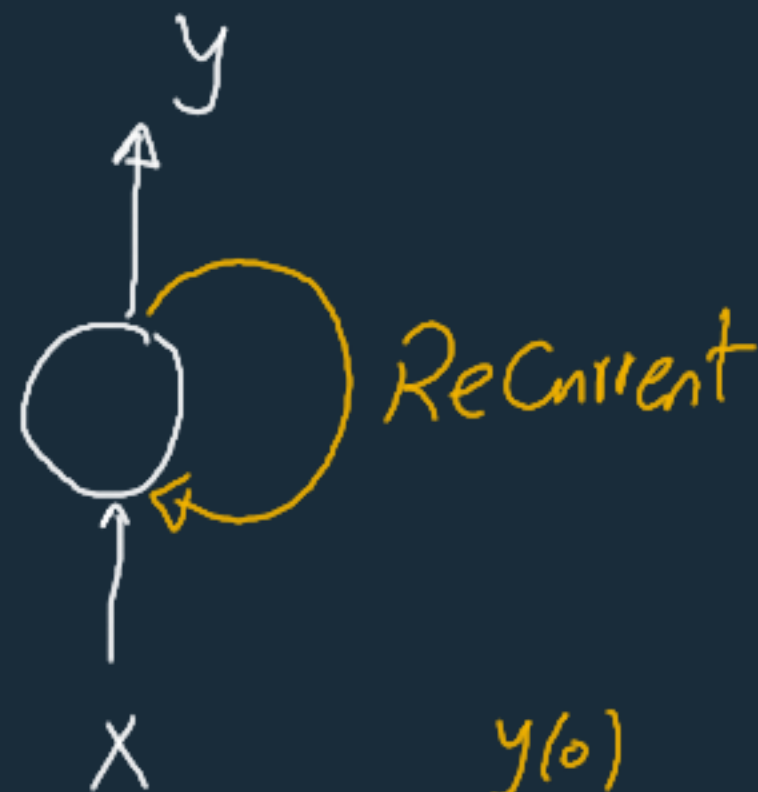


* Text problems with classic ML / ANN Models

- ① Text Semantic (meaning)
- ② No sequence Representation (time series)
- ③ Text problems $\rightarrow$ Custom Models
- ④ Need fixed length

$\Rightarrow$ Recurrent Neural Network (RNN)

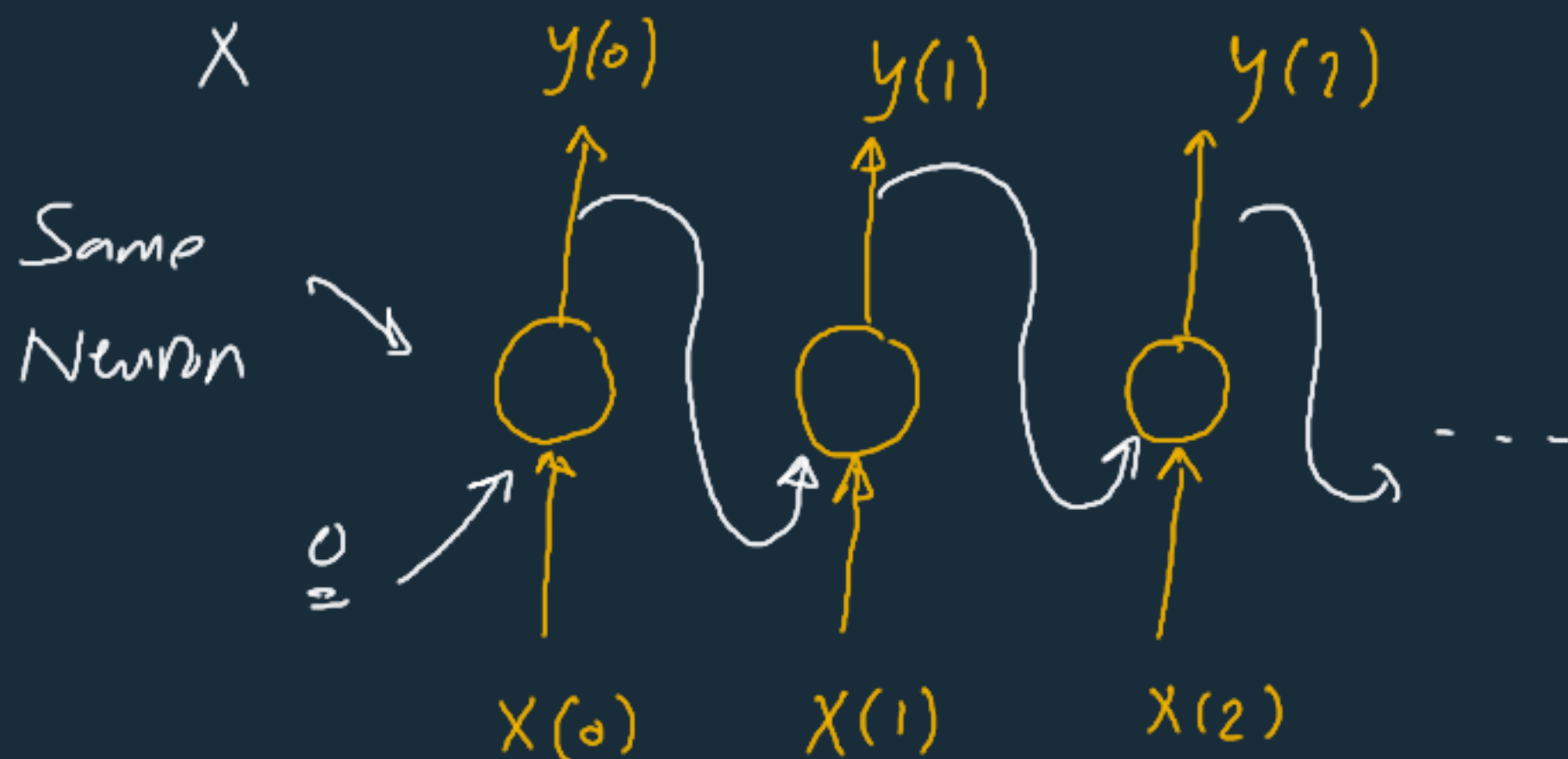
RNN



Current
o/p

$$y(t) = f(x_t, y_{t-1})$$

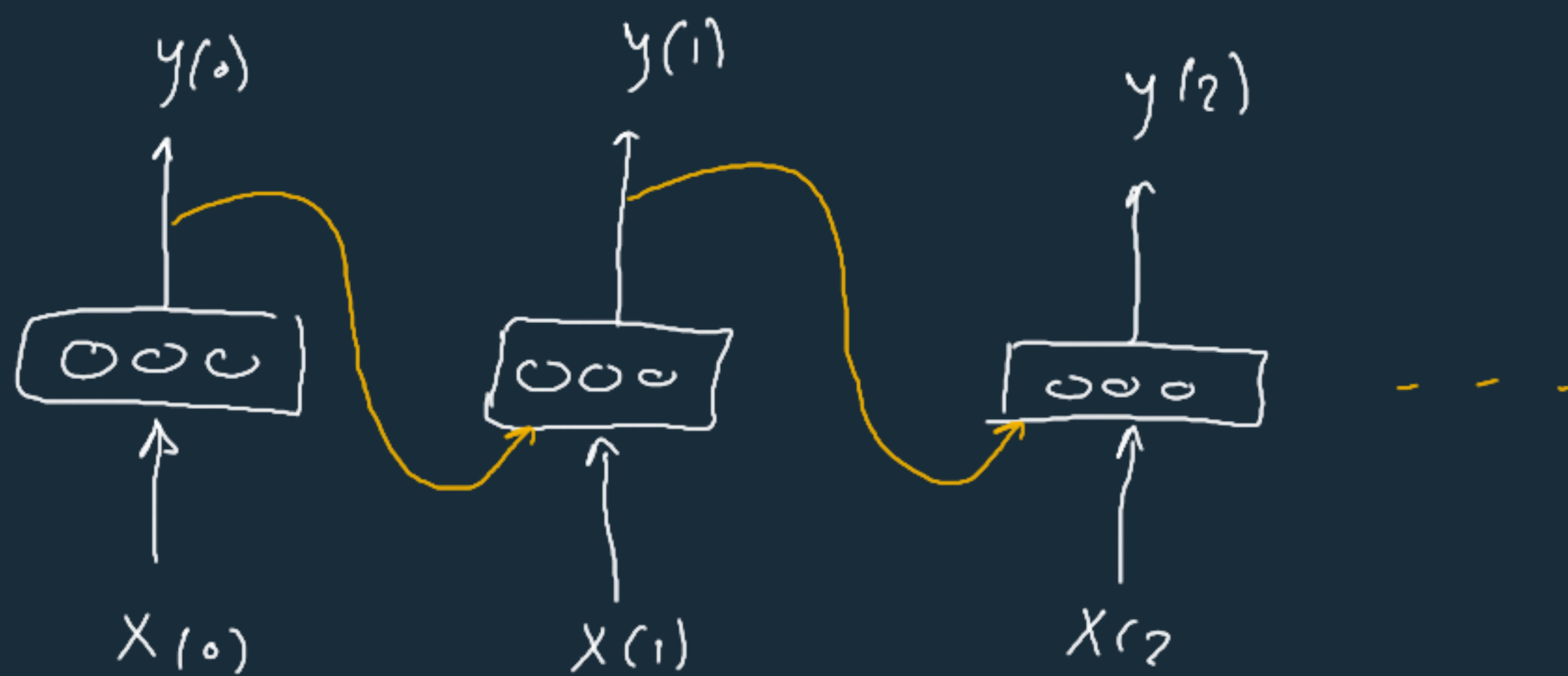
Current I/P prev. o/p



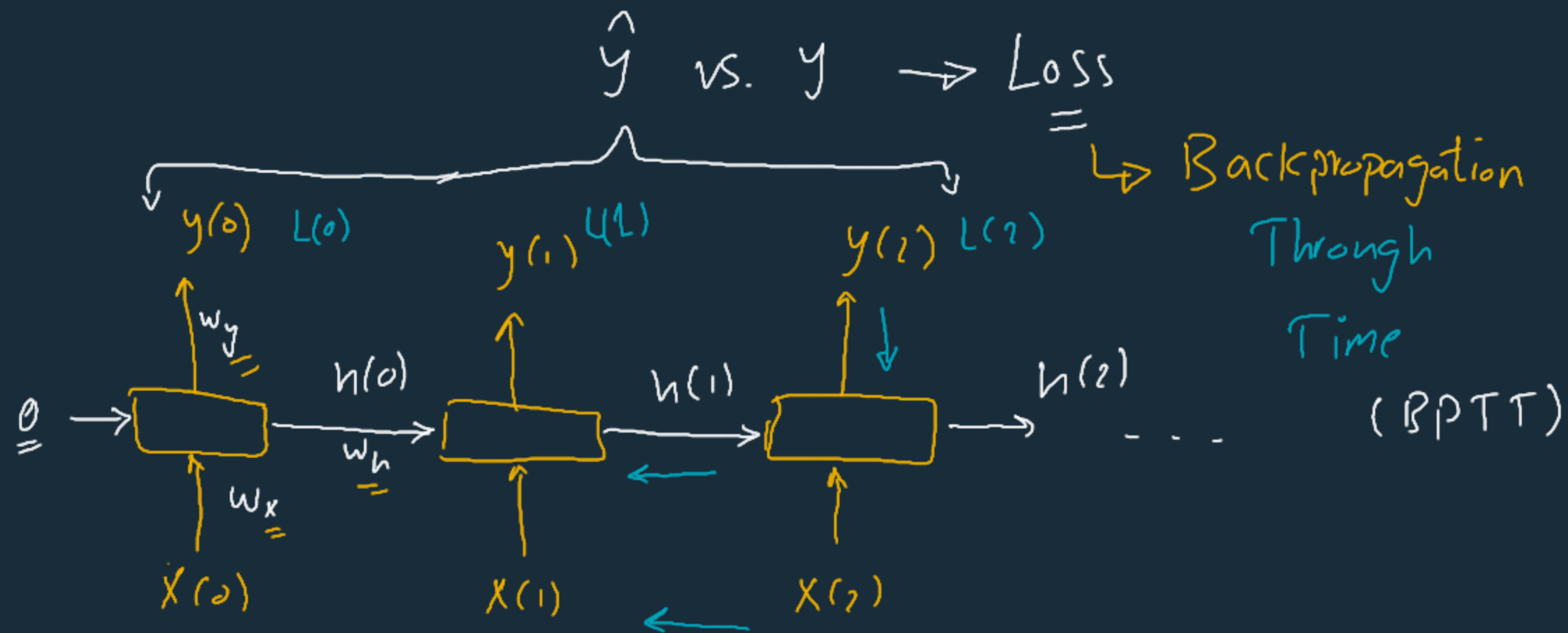
Sequence Representation

Memory Representation

time



Simple RNN



Hidden state:

$$h(t) = \tanh(w_x^T x(t) + w_h^T h(t-1))$$

output:

$$\hat{y}_{(t)} = w_y^T h(t)$$

RNN

ANN $\rightarrow$ No time Representation



RNN $\rightarrow$ Short term Memory

$\rightarrow$ Vanishing gradient

Exploding gradient



New versions of RNN

Save

forget

Control $\leftarrow$ gates



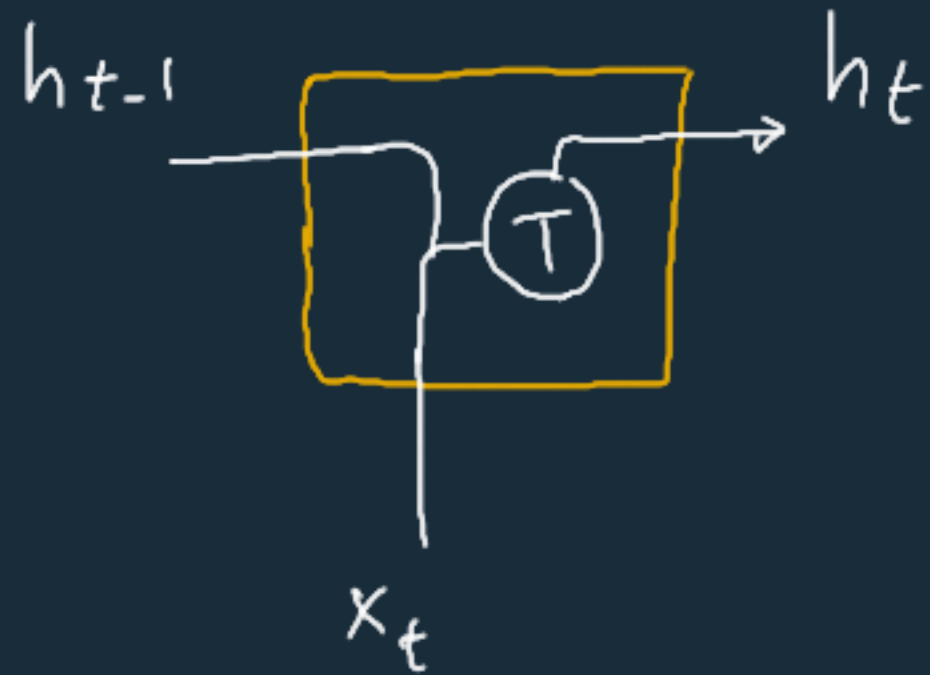
GRU

Gated Recurrent unit

LSTM

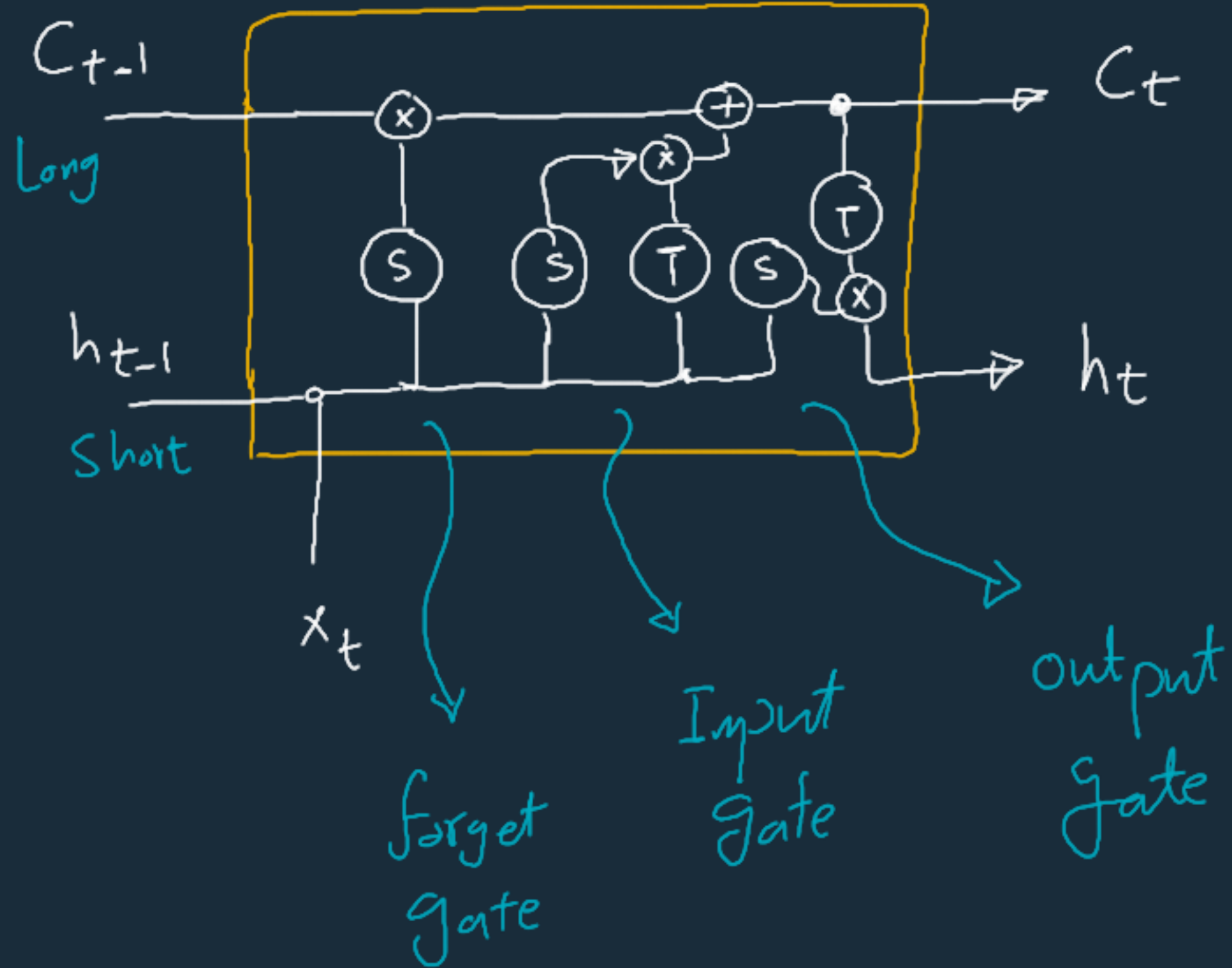
Long short-term Memory

LSTM



RNN

- (S) Sigmoid
- (T) Tanh



RNN



Short Memory

Less Complex

faster

LSTM



Long Memory

More Complex

Slower